

POLICY MEMO - FOR OFFICIAL USE

Why limiting climate change matters for Niger

To: Minister of Environment, Head of Niger’s UNFCCC delegation

From: Your name, Climate Adviser

Date: 14 October 2026

Re: What Niger stands to lose, and to gain if warming is limited

KEY MESSAGE

Niger’s economy rests on the activities that heat and erratic rain hit first: rain-fed farming and herding generate almost 40% of GDP and employ more than 80% of the population. Climate Impact Explorer projections show that at 3 °C of global warming rather than 1.5 °C, the share of Niger’s land hit by a heatwave in a typical year would rise from about ??% to ??%, and the share exposed to crop failure from ??% to ??%. The World Bank estimates that a dry, high-impact climate future could cut Niger’s GDP by up to 11.9% by 2050. **Niger emits a negligible share of global greenhouse gases yet faces some of the largest losses: pushing for a rapid global phase-out of emissions is a matter of national interest, not charity.**

Why Niger is unusually exposed

- ▶ **An open-air economy.** Agriculture and livestock account for almost 40% of GDP and over 80% of jobs, and depend overwhelmingly on rainfall (World Bank, 2024).
- ▶ **Shocks are already frequent and costly.** The 2005 and 2010 droughts triggered national food crises; the 2024 floods killed 339 people and affected more than 1.1 million (AFP, 2024).
- ▶ **Demand is growing fast.** With one of the fastest-growing populations in the world, Niger needs more food, water and jobs every year, from a climate that makes each harder to secure.

Three priority impacts: 1.5 °C versus 3 °C of global warming

Source: Climate Impact Explorer (Climate Analytics), Niger, multi-model median; model range in brackets. A 3 °C world is roughly where current policies lead by 2100; the difference between the two columns is what Niger gains if the world limits warming.

Impact (CIE indicator)	At 1.5 °C	At 3 °C	Why it matters for Niger
1. Extreme heat <i>Land area exposed to heatwaves each year (%)</i>	??% [??–??]	??% [??–??]	Farming, herding and construction are done outdoors, mostly without electricity or cooling. Heat cuts working hours, stresses livestock and raises mortality: GIZ/PIK (2021) project about 50 more very hot days a year and a threefold rise in heat-related deaths by 2080 under high emissions.
2. Crop failure <i>Land area exposed to crop failure each year (%)</i>	??% [??–??]	??% [??–??]	Millet and sorghum feed the country. Past warming has already cut West African millet yields by 10–20% (Sultan et al., 2019), and beyond about +2 °C of local warming no plausible increase in rainfall offsets the heat losses (Sultan et al., 2013). A failed harvest means higher grain prices, malnutrition and distress sales of livestock.
3. River floods <i>Population exposed to river floods each year (%)</i>	??% [??–??]	??% [??–??]	Niamey, the Niger River valley and irrigated rice schemes concentrate people, infrastructure and food production. In 2020 the river reached its highest level since 1929 and paralysed the capital (Al Jazeera, 2020). Niger must prepare for droughts and floods, often in the same decade.

Implications for Niger’s national interest

- 1. The cost is counted in GDP points.** The World Bank (2022) estimates that by 2050 climate change would reduce Niger’s annual GDP by 2.2% in a wet, optimistic scenario but by 11.9% in a dry, pessimistic one, the largest loss among the five G5 Sahel countries. These figures are probably *underestimates*: they leave out migration, conflict and ecosystem degradation.
- 2. The real danger is a cascade, not a single number.** A hot year with erratic rains → failed harvests and pastures → higher grain prices and forced sales of livestock → malnutrition, household debt, migration to cities and farmer–herder tensions → emergency spending that crowds out investment. The warmer the world, the more often these “bad years” occur.
- 3. Adaptation has limits that only global mitigation can push back.** Irrigation, drought-tolerant seeds and crop insurance are essential but costly, and beyond roughly +2 °C of local warming they cannot fully offset heat losses

Save your Climate Impact Explorer screenshot as cie_figure.png

Niger – land area exposed to crop failure – 1.5 °C vs. 3 °C (or 1.5 °C pathway vs. current policies)

Figure 1. Crop-failure exposure in Niger: the gap between the two curves is what limiting warming saves. At 3 °C, about ??% of Niger's land would be exposed to crop failure in a typical year, against ??% at 1.5 °C. The shaded bands show model uncertainty: even the low end of the 3 °C range lies above the high end of the 1.5 °C range, so the direction of the risk is robust. Source: Climate Impact Explorer, Climate Analytics.

(Sultan et al., 2013). Every tenth of a degree avoided lowers an adaptation bill Niger cannot pay alone.

Recommended negotiating position. (i) Defend the 1.5 °C goal and press major emitters for NDCs consistent with a rapid fossil-fuel phase-out; (ii) tie Niger's support to scaled-up, accessible adaptation finance with simplified access for Least Developed Countries; (iii) use the Loss and Damage Fund for droughts and floods, in coalition with the African Group and the LDC group.

What the evidence does, and does not, say

Rising heat is the most robust signal: all models agree on its direction. **For rainfall, models disagree even on the sign of change**, and some project small millet yield gains from CO₂ fertilisation (GIZ/PIK, 2021). The risk to plan for is therefore not a certain fall in average yields but greater variability, with more drought years *and* more flood years. These figures are projections, not forecasts, and they describe hazards only: Niger's high social vulnerability means actual impacts are likely to be larger than the physical numbers suggest.

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